Multi-Direction Networks With Attentional Spectral Prior for Hyperspectral Image Classification

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***Abstract*— Convolutional neural networks (CNNs) have achieved prominent progress in recent years and demonstrated remarkable properties in spectral–spatial hyperspectral image (HSI) classification. However, conventional spatial-context-based CNNs commonly adopt the single patchwise scheme to represent the to-be-classified samples, which often fails to completely investigate the wealthy spectral–spatial information in complicated situations. For instance, it has great probability to cause misclassifications on the irregular or inhomogeneous areas, especially for the borders across different classes. To counteract this deficiency, we propose a unified multi-direction network (MDN) for HSI Classification (HSIC), which can exhaustively explore the abundant spectral and detailed spatial-context information through multi-direction samples. Additionally, considering the image-spectrum merged structure of the HSI, 3-D Squeeze-and-Excitation residual (3DSERes) blocks are devised in each stream of the framework to consecutively learn the spectral and spatial from low-level to high-level features. Specifically, 3DSERes can not only facilitate fluent gradient in backpropagation through skip connections, but also emphasize the significant spectral–spatial features and constrain the futile ones. This characteristic is beneficial to enhance the model's generalization capability even with limited training samples. Furthermore, for properly aggregating the multi-direction deep features, we exploit the simple, yet effective attentional spectral prior (ASP) creatively through leveraging the original spectral correlations. Extensive experimental results on three benchmark data sets indicate that the proposed MDN-ASP can achieve promising classification performance compared to the state-of-the-art methods.**

***Index Terms*— 3-D squeeze-and-excitation residual (3DSERes) blocks, attentional spectral prior (ASP), hyperspectral image classification (HSIC), multi-direction.**

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I. Introduction

HYPERSPECTRAL imagery (HSI) is captured with dozens to hundreds of contiguous and narrow spectral channels containing a wealth of spectrum information, which is nontrivial to identify various materials at a refined pixel level. Thus, as a research hotpot focusing on recognizing each unique pixel, HSI classification (HSIC) has been introduced into many applications, such as mineral exploitation, precision agriculture, military reconnaissance [1], [2], and so on.

In the initial phase of HSIC, many classic supervised machine-learning algorithms were employed to exclusively interpret the data in spectral domain, such as random forest [3], extreme learning machine (ELM) [4], support vector machine (SVM) [5], and so on. However, due to the intrinsic spatial discrepancy of spectral signatures, the classification performance is inferior if only the spectral profile is took into account [6]–[8]. To overcome this limitation, spatial context was naturally introduced, which has been proved to be effective and indispensable [9]–[11]. For instance, extend morphological profiles (EMPs) were presented in [12], which exploited various structural elements to perform on the top several principal components of HSI data and then fed into an SVM classifier to achieve impressive performance. Afterwards, the extended attribute profiles (EAPs) [13], [14] and extinction profiles (EPs) [15] were proposed, which could acquire improved classification results. However, these techniques with manually predefined hyperparameters can only extract the shallow spectral–spatial handcraft features, which are deemed to be under-represented.

Recently, deep learning (DL)-based approaches are developed to effectively learn from low-level to high-level hierarchical features and discriminative representations [16], [17]. Different from the aforementioned traditional methods, DL classification models are performed in an end-to-end data-driven fashion, of which the abundant parameters are directly learned from the input data. The canonical algorithms include deep belief networks (DBNs) [18], recurrent

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neural networks (RNNs) [19], fully convolutional neural networks (FCNs) [20], [21], and convolutional neural networks (CNNs) [22], to name a few. Particularly, due to the benefits of local receptive filed and sharing weights, CNNs have been developed into the most prevalent method in computer vision and demonstrated encouraging performance in HSIC. Typically, 1D-CNN [23], 2D-CNN [24], and 3D-CNN [25] were successively applied to exploit the spectral, spatial, and spectral–spatial information, respectively, which obtained promising classification results. However, in such an automatic data-driven learning manner, these naive CNN models face the challenge of over-fitting when processing the high-dimensional HSI data under the condition of insufficient training samples [26]. And it is notable that labeling is relatively expensive, laborious, and time-consuming in practical applications [27], [28].

To counteract this ill-posed problem, Li *et al.* [29] proposed to involve pixel-pair features into CNNs (PPF-CNN) through coupling the available labeled pixels, and thus the training samples were largely augmented to gain satisfactory test accuracy. On the other hand, reducing the data dimensionality and extracting more representative and discriminative spectral–spatial features can also mitigate the over-fitting phenomenon to some degree [30]–[33]. Intuitively, based on the fact that neighboring pixels are prone to be the same land-cover type, patch-wise scheme is commonly adopted to integrate the spectral–spatial information. In this way, each pixel to-be-classified is assigned to the center of a fixed-size square neighborhood, which can force to produce advanced classification result. For instance, Zhong *et al.* [34] proposed a 3-D spectral–spatial residual network (SSRN) to successively exploit the spectral and spatial features. Owing to the residual block, SSRNs can lower the convergence loss and alleviate the attenuation of classification accuracy as the networks developing deeper. Then, Wang *et al.* [35] presented a fast dense spectral–spatial convolutional network (FDSSCN) for HSIC, which acquired satisfactory accuracy with high efficiency through introducing dense connection into the model. Additionally, in order to highlight informative features and restrict the trivial ones, attention mechanism has been embedded into various HSIC frameworks [36]–[38]. For example, Fang *et al.* [40] employed the Squeeze-and-Excitation network (SENet) [39] to learn attention paying on the spectral domain, which made the spectral dimension more discriminative. In [41], a novel hybrid residual attention module was proposed to distill more distinguishing spatial and spectral representations, which further strengthened the scalability of the deep model.

Nevertheless, the aforementioned patch-wise classification framework can only take fixed-size patches as input, which has two drawbacks: First, the pixels included in one fixed-size neighborhood may not belong to the same category, especially for the boundary pixels [42]. Consequently, it will bring bias to the classifier and lead to misclassifications. Second, the deep networks can only extract the local spectral–spatial features constrained by the predefined window size, which is not able to fully exploit the spatial context information around a specific pixel. To alleviate these problems, many studies propose to utilize multiscale patches to exploit the compensate spatial information in a multiple input style during the process of feature extraction [43]–[45]. In particular, Li *et al.* presented a diverse-region CNN (DR-CNN) to take advantage of the rich spectral–spatial information, which achieved impressive performance [46]. However, DR-CNN needs to design different networks for diverse patch inputs, and it still focuses on exploiting the local information restricted in the window size. Additionally, the above methods [43]–[46] with multi-tunnel architectures simply stack the extracted deep features together to ratiocinate the final result, ignoring the different significance of different patches or regions centered on the same target pixel.

To conquer these challenges, we propose an innovative multi-direction network (MDN) in this study. In the MDN, the input of the network exactly includes nine directional regions (i.e., central, left, right, up, bottom, upleft, upright, bottomleft, and bottomright) oriented by the pixel of interest (PoI). In this manner, the framework is extended to be wider, liberating the scope of perception from the constraints of local window to catch a longer range of spatial dependence. In order to extract more representative features and ensure the network convergence, we develop a 3-D Squeeze-and-Excitation residual (3DSERes) block by incorporating 3DSENet into residual blocks to consecutively exploit the spectral and spatial information in each channel. Different from DR-CNN, since the size of multi-direction patches are in accordance with each other, they can share the same network architecture in all streams of the MDN. Moreover, first, we believe that the original natural spectral information can be reused except for extracting deep representations from them, because it contains the most important information to distinguish different materials in HSI [8]. Second, it is intuitive that each direction patch belonging to the same PoI should have a different degree of importance for reasoning the prediction. Thus, we embed the attentional spectral prior (ASP) into the multi-stream networks leveraging the primitive spectral correlations, of which the compensatory information is favorable for precisely aggregating the multi-direction deep features. Then, a softmax-based classifier is adopted to determine the label of each pixel.

In summary, the primary contributions of this work are specified as follows.

1) An innovative MDN is proposed for HSIC, which is able to take advantage of multi-direction contextual features for each individual PoI. Consequently, the abundant spectral and detailed spatial-context information is thoroughly and precisely exploited, so as to accurately classify the samples, especially for the boundary pixels across categories.
2) To further boost the classification performance, in each stream of the MDN, we devise successive 3-D Squeeze-and-Excitation residual (3DSERes) blocks through incorporating the 3DSENet into residual architectures. This structure can not only facilitate fluent gradient in backpropagation through skip connections, but also model the codependency of the feature maps and calculate their different importance by utilizing the global spatial contextual information.

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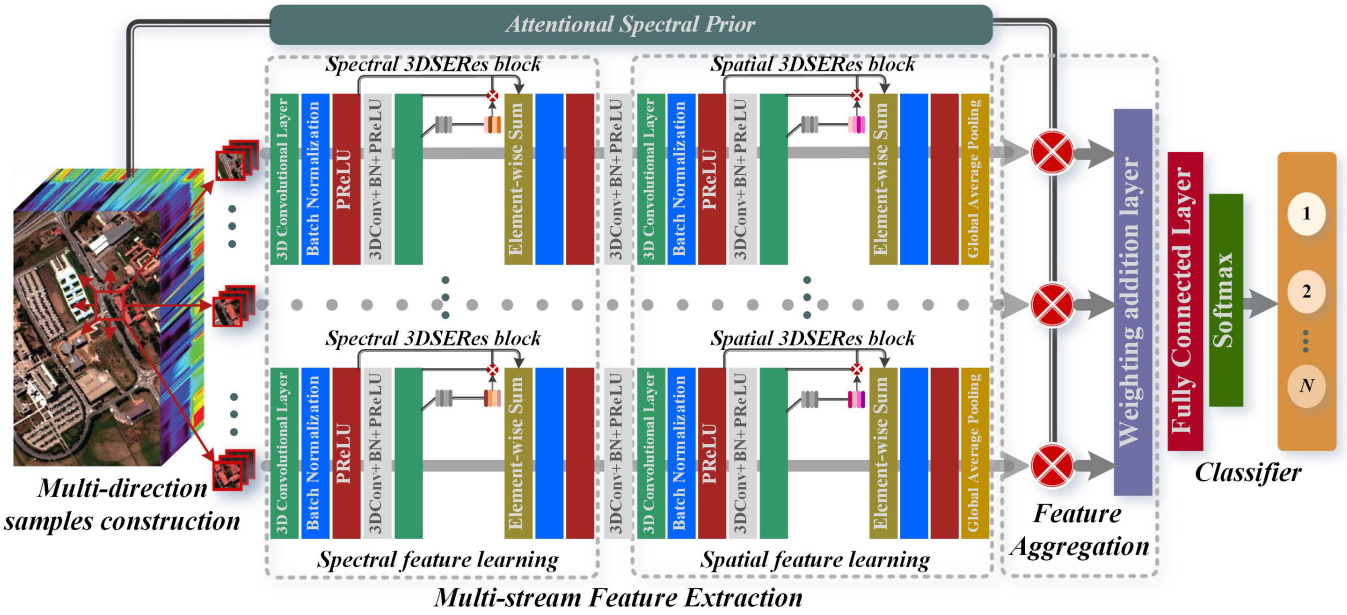

Fig. 1. Framework of our proposed MDN-ASP for HSIC. It is composed of four components: multi-direction samples construction, multi-stream feature extraction, feature aggregation with ASP, and a softmax-based classifier. The same color represents the layers with same operation.

3) Leveraging the original spectral relationships, we involve the simple yet effective ASP into the proposed MDN to weighted aggregate the extracted multi-direction deep spectral–spatial features, which is observed to be more appropriate than the assembly approach without distinction.
4) The ingenious architecture design enables the MDN-ASP a unified framework that can be expediently generalized to three commonly studied HSI benchmarks. Particularly, the MDN-ASP surpasses the compared state-of-the-art HSIC approaches and achieves encouraging classification accuracy with limited training data.

The rest of this article is organized as follows. Section II depicts the proposed framework at length. Section III carries out abundant experiments and gives detailed analysis of the classification results. Section IV draws the conclusions.

II. Proposed Method

A. Architecture of the Proposed MDN-ASP

The framework diagram of the proposed MDN-ASP for HSIC is delineated in Fig. 1, of which the backbone comprises multiple paralleled branches to investigate the input multi-direction patches. We can observe that it mainly consists of four components: multi-direction samples construction, multi-stream feature extraction, feature aggregation with ASP, and a softmax-based classifier. First of all, the multi-direction samples are generated incorporating the target pixel with a removable window. When receiving these samples, each sub-network exploits two continuous 3DSERes blocks to separately learn discriminative spectral and spatial representations. After that, the multi-direction deep features are aggregated by a weighting addition layer, which is completed relying on the ASP. Finally, the fused feature is fed into the last fully connected layer and softmax activation function to derive the probability vector indicating the target pixel's category. The networks of feature extraction and classifier are trained in an end-to-end manner without any data preprocessing techniques. Next, each component of the framework will be elaborated in detail.

B. Multi-Direction Samples Construction

Different from the commonly used single-patch-based classification model, we employ the direction information from the pixel of interest (PoI) to construct multi-direction samples. Given an HSI data set denoted as $\mathbf{X} \in R^{H\times W\times V}$, where H and W represent the height and width of the spatial dimension, respectively, and V refers to the amounts of spectral bands. Then, the labeled samples can be represented as $\mathbf{S}=\{\mathbf{s}_i\}_{i=1}^{l} \in R^{V\times l}$ and the corresponding label y_i belongs to $\{1, 2, \ldots, N\}$, where N is the class number. For an arbitrary to-be-classified PoI denoted as $\mathbf{s}$, we first employ a $P \times P$ window to map its center with $\mathbf{s}$ for extracting the central patch $\mathbf{s}_c \in R^{P\times P\times V}$. Then, the window is moved toward eight different directions until the PoI reaches the edges or corners. Consequently, the other eight direction patches are produced, that is, $\mathbf{s}_{\mathrm{l}}$, $\mathbf{s}_{\mathrm{r}}$, $\mathbf{s}_{\mathrm{u}}$, $\mathbf{s}_{\mathrm{b}}$, $\mathbf{s}_{\mathrm{ul}}$, $\mathbf{s}_{\mathrm{ur}}$, $\mathbf{s}_{\mathrm{bl}}$, and $\mathbf{s}_{\mathrm{br}}$ representing the left, right, up, bottom, upleft, upright, bottomleft, and bottomright patches, respectively. From another point of view, $(2P-1)\times(2P-1)$ (i.e., 5×5 when $P=3$) scope around the PoI are cropped to subtly involve abundant spectral–spatial information into the multi-directions patches.

Illustratively, the procedure of multi-direction samples construction is depicted in Fig. 2 when the neighborhood size $P=3$. From Fig. 2, we can find that the benefits of the proposed multi-direction samples mainly lie in two aspects:

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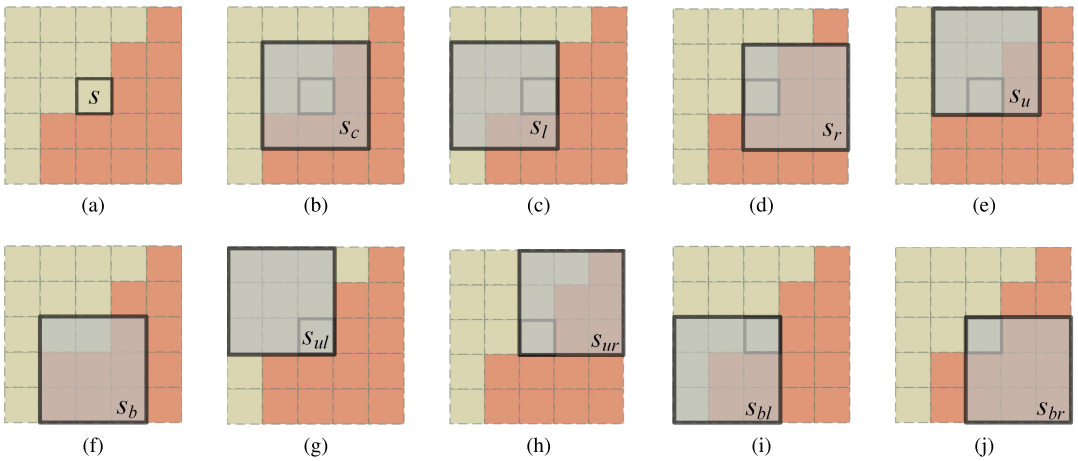

Fig. 2. Illustration of a sample with multi-direction (exactly nine directions) when the window size equals to 3. (a) Center pixel. (b) Central. (c) Left. (d) Right. (e) Up. (f) Bottom. (g) Upleft. (h) Upright. (i) Bottomleft. (j) Bottomright.

First, the proposed multi-direction scheme can catch larger scope of spatial dependence out of the initial fixed window size. Second, it can be found that the central patch $\mathbf{s}_c$ of the border pixel $\mathbf{s}$ contains materials from two classes. If we only employ this single patch, it has great probability to predict the label to the incorrect category, since the patch has more pixels belonging to the wrong class than the right one. By contrast, when using the multi-direction strategy, there is at least one patch only includes the same object with the border pixel (i.e., upleft patch $\mathbf{s}_{ul}$) in the given circumstance, which can facilitate to predict the boundary pixels more accurately.

C. Multi-Stream Feature Extraction

For the sake of entirely investigating the abundant spectral information and complementary spatial context contained in the multi-direction samples, we design a multi-stream network to excavate the high-level representative features. Since the input volume of each stream is in equal size, the subnetworks can share the same architecture in the procedure of feature extraction. Considering HSI is comprised of one spectral and two spatial dimensions, our unified framework is designed to consecutively learn the spectral and spatial features. This means that in the first spectral feature learning stage, the kernels of the 3-D convolution (3DConv) [25] in the spatial dimension are assigned as 1*1 and the convolutional operation is primarily performed on the spectral dimension. By contrast, in the following spatial feature learning phase, the 3DConv kernels in the spectral dimension are set to 1, and the convolutional calculation is mainly imposed on the spatial dimension. Additionally, in each stream, 3DConv layers combined with batch normalization (BN) [47] and PReLU nonlinear activation function [48] are assembled as the basic building blocks, which can make the training process more efficient and ensure faster convergence speed of the model. Although CNNs have demonstrated powerful ability to extract discriminative features due to the sparsity constraint, the loss value calculated at different level of activation layers may subject to the problem of vanishing gradient [34]. As the networks designing deep, the classification performance will be greatly decreased, especially for training with insufficient labeled samples. To resolve this defect, we add skip connections to every other convolutional layers in both the spectral and spatial feature learning procedures, which indeed formulate the hierarchical representations into the effective residual architectures [49]. In this way, the computed gradient will be transmitted from the deeper layer to the former ones and the network will converge in a smaller loss value. Moreover, to further boost the learning ability of the spectral–spatial networks, we embed 3-D Squeeze-and-Excitation (3DSE) module into the residual block, which is referred to 3DSERes block in this article. It is worth mentioning that the 3DSE module can leverage the holistic spatial context through global pooling to model the relationships of non-local feature maps. This means that the informative feature maps are enhanced while the nonsignificant ones are suppressed, which are conducive to adequately explore the superfluous spectral bands and global spatial contextual information. Next, we will describe the 3DSERes block-based spectral and spatial feature extraction networks in detail.

As shown in Fig. 1, suppose that the predefined kernel shape of the ith 3DConv layer is $(h, w, d, c_{\text{in}}, c_{\text{out}})$, where (h, w, d) represent the sizes along the height, width, and depth dimensions, respectively. c_{in} denotes the amounts of input channels, and c_{out} refers to the 3-D filter number which equals to the output channels. Then the 3DConv operation can be formulated as follows:

$$v_{ij}^{xyz} = \text{act}\left(b_{ij} + \sum_{c}\sum_{\alpha=0}^{h_i-1}\sum_{\beta=0}^{w_i-1}\sum_{\gamma=0}^{d_i-1} k_{ijc}^{\alpha\beta\gamma} \times v_{(i-1)c}^{(x+\alpha)(y+\beta)(z+\gamma)}\right) \tag{1}$$

where v equals to the feature map value and (x, y, z) represent the location in the jth feature cube of the ith layer. act$(\cdot)$ refers to the activation function and b represents the bias. c indexes past the staff of feature cubes in the $(i-1)$th layer, and (α, β, γ) are indexes of the convolutional filters. To be specific, the filter sizes of the first convolutional layer in the

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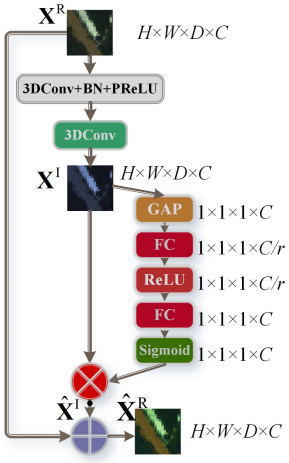

Fig. 3. Implementation of 3-D Squeeze-and-Excitation Residual (3DSERes) block.

spectral and spatial feature learning procedure are assigned to $(1, 1, 7, V,\ C)$ and $(3, 3, C, C, C)$, respectively. With regard to the activation function, we choose PReLU [48] for speeding up the training process, which can be formulated as follows:

$$g(x_i) = \begin{cases} x_i, & x_i \geq 0 \\ u_i x_i, & x_i < 0 \end{cases} \tag{2}$$

where x_i is the input of the nonlinear activation on the ith channel. u_i determines the negative part's slope of the activation value, which is a learnable parameter that can be updated by the momentum method.

$$\nabla u_i = a_i \nabla u_i + \gamma \frac{\partial \varepsilon}{\partial u_i} \tag{3}$$

where a_i is the momentum and γ is the learning rate. The initial value of u_i is assigned as 0.25. Besides, in order to further accelerate the model convergence, batch normalization (BN) transform is employed just before the PReLU function, which is calculated as follows:

$$\widehat{\mathbf{X}}^i = \frac{\mathbf{X}^i - E\left[\mathbf{X}^i\right]}{\sqrt{\mathrm{Var}\left[\mathbf{X}^i\right]}} \tag{4}$$

where $\mathbf{X}^i$ represents the feature input from the ith layer. $E[\cdot]$ and Var$[\cdot]$ are the operations obtaining expectation and variance, respectively. $\widehat{\mathbf{X}}^i$ denotes the normalized counterpart.

In addition to fundamental operations of 3DConv, BN, and PReLU, the feature learning process includes two 3DSERes blocks to highlight the crucial channel-wise spectral and spatial features contained in the multi-direction samples. Concretely, the architecture of 3DSERes block is completed in accordance with Fig. 3. Assume that there are C feature cubes with the shape of $H \times W \times D$, denoted by $\mathbf{X}^R \in \mathbf{R}^{H\times W\times D\times C}$ in the $(l-1)$th layer, where H, W, and D are the height, width, and depth of each feature cube, respectively. It is notable that $D > 1$ and $D = 1$ in the spectral and spatial 3DSERes blocks, respectively. Next, the interlayer $\mathbf{X}^I$ is derived through sequential 3DConv, BN, PReLU, and 3DConv calculations. In more specific terms, the filter sizes of these two 3DConv are $(1, 1, 7, C, C)$ in the spectral 3DSERes block and $(3, 3, 1, C, C)$ in the spatial 3DSERes block. Afterwards, the approach to gain the channel-wise attention factors is described as follows.

At first, we compress the entire spatial information into a channel descriptor by using 3-D Global Average Pooling (GAP) operation. This means that the channel-wise statistic $\mathbf{S} \in \mathbf{R}^{1\times1\times1\times C}$ is acquired by

$$\mathbf{S}_c = \frac{1}{H \times W \times D} \sum_{x=1}^{H} \sum_{y=1}^{W} \sum_{z=1}^{D} \mathbf{X}_c^I(x, y, z) \tag{5}$$

where $c = \{1, 2, \ldots, C\}$ and $\mathbf{X}_c^I(x, y, z)$ is the element value at position (x, y, z) of cth feature cube in $\mathbf{X}^I$. Then, the parameters are learned to explicitly imitate the correlation between the feature channels. To be more concrete, $\mathbf{S}$ is passed to the key excitation transform to get the attention weight vector $\mathbf{A}_{tt} \in \mathbf{R}^{1\times1\times1\times C}$.

$$\mathbf{A}_{tt} = \Phi(\mathbf{W}_U \mathrm{act}(\mathbf{W}_D \mathbf{S})) \tag{6}$$

where act is the ReLU nonlinear activation function [50]. $\mathbf{W}_D$ and $\mathbf{W}_U$ are the weights of the first and second fully connected (FC) layers, which behave as downscaling and upscaling with the identical reduction and increase ratio value r, respectively. These two FCs make up a bottleneck structure, which has more nonlinearity for better fitting the complex correlation between channels. It can also greatly reduce the amounts of parameters than using only one FC layer. $\Phi(\cdot)$ denotes the sigmoid function, which acts as a gate to constrain the attention vector values in the (0, 1) scope. Subsequently, broadcasted element-wise multiplication denoted as $\otimes$ is executed to obtain the channel statistics of $\hat{\mathbf{X}}^I$.

$$\hat{\mathbf{X}}_c^I = \mathbf{X}_c^I \otimes \mathbf{a}_c \tag{7}$$

where $\mathbf{a}_c$ is the cth element of attention score $\mathbf{A}_{tt}$. Finally, we have the output of the 3DSERes block represented by

$$\hat{\mathbf{X}}^R = \mathbf{X}^R + \hat{\mathbf{X}}^I. \tag{8}$$

D. Feature Aggregation With ASP

Various materials have different reflective properties, so intra-class spectral similarity and inter-class diversity are one of the fundamental assumptions in HSIC, which enables the pixel-level categorization a possible task. On the other hand, for a given PoI, it is intuitive that the samples generated from different directions should have different importance to facilitate the network making a more accurate decision. Taking Fig. 2 as an example, we should pay more attention to the left, up, and upleft directions, which contain more pixels same as the PoI. In contrast, the right, bottom, and bottomright samples should be suppressed, which may lead misclassification to the final decision. Based on these knowledge, we incorporate the simple yet effective ASP into the networks to flexibly aggregate the extracted multi-direction deep features.

Concretely, leveraging the prior spectral information, we can calculate the attention score of the ASP for the dth direction by

$$a_d = e^{-\alpha \left\| \mathbf{s}_d^{\mathrm{aver}} - \mathbf{s} \right\|_2^2} \tag{9}$$

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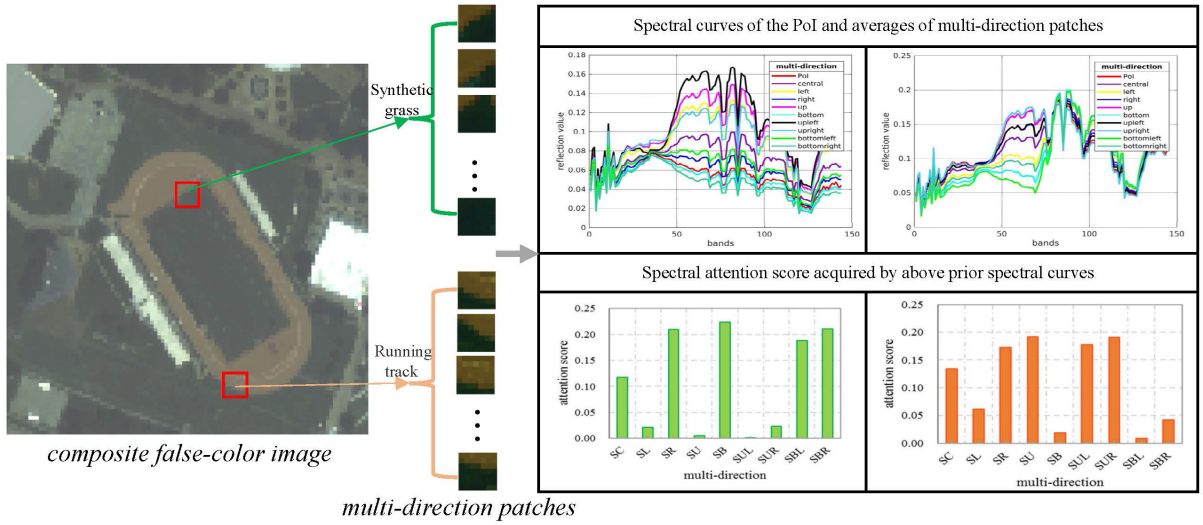


Fig. 4. Schematic diagram declaring the motivation and effectiveness of the ASP.

where $\mathbf{s}_d^{\text{aver}}$ represents the average value of the dth direction sample along the spatial dimension and $\mathbf{s}$ is the PoI. $\|\cdot\|_2^2$ refers to the square of the $l2$ norm and α is a steepness parameter controlling the gap of the attention scores. Then, the attention scores are normalized from 0 to 1 and injected into the feature aggregation process, which is completed through a weighting addition layer formulated as

$$f = \sum_{d=1} f_d \otimes a_d \tag{10}$$

where f_d is the extracted deep feature from the dth direction.

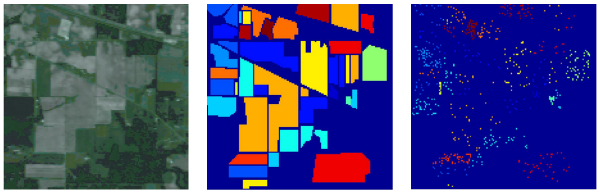
Fig. 4 further declares the motivation and effectiveness of the ASP, where the first row plots the spectral curves of the PoI and averages of the multi-direction patches, while the second row shows the normalized spectral attention score acquired by the prior spectral relationships. From Fig. 4, it is physically interpretable that the more pixels of same category as the PoI in a patch, the greater importance (i.e., attention score) is in feature aggregation, which is indirectly advantageous for accurate classification, notably for the boundary pixels.

E. Softmax-Based Classifier

At the end of the proposed networks, the attentional merged feature f is fed into a fully connected layer and softmax-based classifier to obtain the predicted probabilities. For this kind of end-to-end framework, we employ the commonly used cross-entropy penalty function to serve as the optimization object. Mathematically, it is formulated as follows:

$$J = \min_{\theta} \left(-\sum_{i=1}^{bs} \log \frac{e^{\mathbf{W}_{y_i}^T f_i + b_{y_i}}}{\sum_{j=1}^{N} e^{\mathbf{W}_j^T f_j + b_j}} \right) \tag{11}$$

where θ represents the trainable parameters of the networks and bs means the batchsize. f_i denotes the merged feature belonging to the y_ith class. $\mathbf{W}_j$ is the jth column of the weight $\mathbf{W}$ in the fully connected layer, and b refers to the bias.



(a) (b) (c)

Fig. 5. IP data set. (a) False-color image (18th, 25th, and 30th channels). (b) Groundtruth. (c) Standard training samples.

III. Experiments and Analysis

In our simulations, the superiority of the proposed framework is verified with three real-world HSIC benchmarks, that is, Indian Pines (IP), University of Pavia (UP), and University of Houston 2013 (UH2013). We adopt several classic and state-of-the-art HSIC methods as the comparisons, such as SVM [5], 2D-CNN [24], 3D-CNN [25], PPF-CNN [29], SSRN [34], FDSSCN [35], and DR-CNN [46]. Additionally, all of the experiments are performed on a computer with Intel Xeon E5-1620 v4 processor and an NVIDIA GeForce RTX 2080 GPU. MATLAB and Python 3.6 are used as programming languages. The performance of our proposed classification network is quantitatively assessed by three commonly used criteria, which are overall accuracy (OA), average accuracy (AA), and Kappa coefficient (k).

A. Data Sets Description

The first IP data set is collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Northwestern Indiana, which comprises 145 × 145 pixels with a relatively low geometric resolution as 20 m. The image originally has 220 bands ranging from 0.4 to 2.5 μm. In our study, only 200 spectral bands are remained after removing the water absorption and noisy ones. The false-color image and its groundtruth are exhibited in Fig. 5(a) and (b), respectively.

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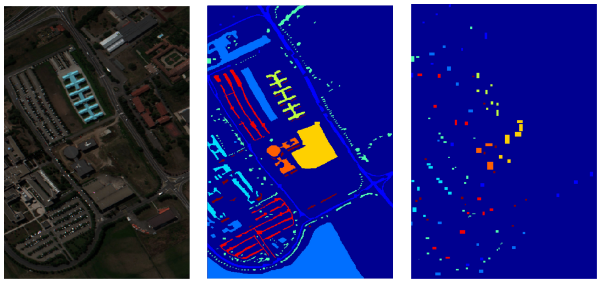
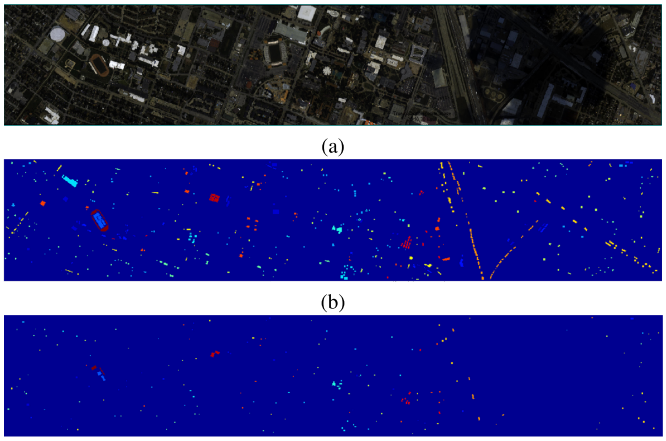


(a) (b) (c)

Fig. 6. UP data set. (a) False-color image (20th, 30th, and 50th channels). (b) Groundtruth. (c) Standard training samples.



(a) (b) (c)

Fig. 7. UH2013 data set. (a) False-color image (20th, 50th, and 70th channels). (b) Groundtruth. (c) Standard training samples. Best zoomed-in view.

TABLE I

LABELED LAND-COVER TYPES AND TRAINING/TEST SAMPLE NUMBERS IN IP DATA SET

No.	Color	Land-cover	Training	Test
1		Alfalfa	15	39
2		Corn-notill	50	1384
3		Corn-mintill	50	784
4		Corn	50	184
5		Grass-pasture	50	447
6		Grass-trees	50	697
7		Grass-pasture-mowed	15	11
8		Hay-windrowed	50	439
9		Oats	15	5
10		Soybean-notill	50	918
11		Soybean-mintill	50	2418
12		Soybean-clean	50	564
13		Wheat	50	162
14		Woods	50	1244
15		Buildings-Grass-Trees-Drives	50	330
16		Stone-Steel-Towers	50	45
-	-	Total	695	9671

We can find there are 16 land-cover classes, of which the specific training and test sample numbers are listed in Table I. It is observed that the annotated sample numbers are imbalance and even in very small size for some classes, which makes IP a challenging yet widely used data set to develop advanced classification models.

TABLE II

LABELED LAND-COVER TYPES AND TRAINING/TEST SAMPLE NUMBERS IN UP DATA SET

No.	Color	Land-cover	Training	Test
1		Asphalt	548	6304
2		Meadows	540	18146
3		Gravel	392	1815
4		Trees	524	2912
5		Painted metal sheets	265	1113
6		Bare Soil	532	4572
7		Bitumen	375	981
8		Self-Blocking Bricks	514	3364
9		Shadows	231	795
-	-	Total	3921	40002

TABLE III

LABELED LAND-COVER TYPES AND TRAINING/TEST SAMPLE NUMBERS IN UH2013 DATA SET

No.	Color	Land-cover	Training	Test
1		Grass Healthy	198	1053
2		Grass Stressed	190	1064
3		Grass Synthetic	192	505
4		Tree	188	1056
5		Soil	186	1056
6		Water	182	143
7		Residential	196	1072
8		Commercial	191	1053
9		Road	193	1059
10		Highway	191	1036
11		Railway	181	1054
12		Parking Lot 1	192	1041
13		Parking Lot 2	184	285
14		Tennis Court	181	247
15		Running Track	187	473
-	-	Total	2832	12197

The second UP data set gathered by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor mainly covers an urban scene. There are 103 bands left in the wavelength scope of 0.43–0.86 μm after pruning the noisy channels. The spatial resolution of this image is 1.3 m. It contains 610 $\times$ 340 pixels, in which 43 923 samples belonging to nine classes are labeled. Accordingly, its false-color image and groundtruth are drawn in Fig. 6(a) and (b), respectively.

The last UH2013 data set is made up of 144 spectral channels in a range from 0.38 to 1.05 μm, which is captured by the Compact Airborne Spectrographic Imager (CASI) in June 23, 2012. The image comprises 349 $\times$ 1905 pixels with a spatial resolution of 2.5 m. As seen in Fig. 7(a) and (b), 15 labeled land-cover types exist in this image. Besides, this public data set is also called grss_dfc_2013, since it was first open-sourced by the Image Analysis and Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) in the 2013 Data Fusion Contest.[1]

To evaluate the proposed framework, the training samples are artificially chosen from the given groundtruth, of which the positions are marked in Figs. 5(c)–7(c), and the corresponding numbers of training and test sample splits are reported from Tables I–III. In recent studies, this standard fixed sampling manner has been wildly used as benchmarks [10], [51], [52], of which the advantages reflect in two aspects. First, compared to the simple random sampling strategy that usually produces

[1]http://www.grss-ieee.org/community/technical-committees/data-fusion/

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over optimistic classification accuracies, such a challenging context can force to estimate the potential and performance of the extracted representations more effectively. Second, utilizing this uniform training and test split to take experiments on various algorithms ensures their fair comparison, which in turn contributes to attaining convincing and reliable conclusions.

B. Experimental Configurations

With the objective to make best use of the proposed framework and demonstrate its effectiveness persuasively, we design elaborate experimental settings, which are delineated in the following.

First, for all experimental data sets, we randomly select 10% samples of the training data to serve as the validation set for developing the networks. This means that 90% of the training samples listed in Tables I–III are employed to learn the optimal kernel weights and bias through back-propagation mechanism while the remained 10% are used to guide the adjustment of hyperparameters, such as the learning rate, batchsize, and so on. Next, we arrange the $P \times P \times V$ patch samples to incorporate the spectral–spatial statistics according to the training and test positions and then generate the multi-direction samples using the strategy depicted in Section II-B. After that, the attention scores for all labeled samples of each direction are calculated using (9) and stored for the subsequent weighting addition layer completed by (10). Concretely, we empirically set the window size $P = 7$ and its influence will be analyzed in Section IV-E. The steepness parameter α in (9) is set to 1. According to Fig. 1, all data sets share the same unified network architecture, of which the backbone has nine branches with the same constitution and the feature cube number denoted as C is experimentally set to 32. For the sake of rebuilding the frameworks, taking UP data set as an example, Table IV shows the detailed data flow of the subnetwork before feature aggregation.

Except for the input samples construction and network architecture design, there are two other significant hyperparameters need to be validated to ensure the model's appropriate convergence, that is, learning rate and training epochs. In specific, first, the weight update magnitude depends on the initialized learning rate. If the setting is too large, the result will get over the optimal value, and if it is too small, the convergence speed will be too slow. Second, the fitting degree of the deep model is impacted by the epoch number, which means that small epoch number may lead to under-fitting while a large one may result in over-fitting phenomenon. Thus, the grid search method is carried out to earn the optimal learning rate from $\{1 \times e^{-2}, 1\times e^{-3}, 1\times e^{-4}, 1\times e^{-5}\}$ and epoch number from $\{50, 100, 150, 200\}$. Here, Adam stochastic optimization method is utilized to better handle the sparse gradient problems. Then, according to the validation accuracies varying with the them, we can confirm the best learning rate is $1 \times e^{-4}$ and the best epoch number is 100. The small value of epoch number also implies rapid convergence of our proposed rational model. Besides, the decay factor of the learning rate is assigned to $1 \times e^{-6}$ and the batchsize in the experiment is set to 16. With regard to the reduction ratio r in the 3DSERes module, it is experimentally validated to be 8.

TABLE IV
DETAILED DATA-FLOW OF THE DESIGNED SUBNETWORK BEFORE FEATURE AGGREGATION FOR UP DATA SET

Layers	Input size	Output size	Kernel size
3DConv+BN+PReLU	(7, 7, 103, 1)	(7, 7, 49, 32)	(1, 1, 7, 32)
Spectral 3DSERes block	(7, 7, 49, 32)	(7, 7, 49, 32)	**apply Fig. 3**
BN_PReLU	(7, 7, 49, 32)	(7, 7, 49, 32)	—
3DConv+BN+PReLU	(7, 7, 49, 32)	(7, 7, 1, 32)	(1, 1, 49, 32)
3DConv+BN+PReLU	(7, 7, 32, 1)	(5, 5, 1, 32)	(3, 3, 32, 32)
Spatial 3DSERes block	(5, 5, 1, 32)	(5, 5, 1, 32)	**apply Fig. 3**
BN_PReLU	(5, 5, 1, 32)	(5, 5, 1, 32)	—
Global Average Pooling	(5, 5, 1, 32)	(1, 1, 1, 32)	—

C. Ablation Studies

From previous analysis of our proposed framework, we can recall that its novel components include multi-direction samples construction, 3DSE module cooperation, and ASP integration. Thus, with the purpose of accessing their respective effectiveness, ablation experiments are conducted on all experiment data sets. Specifically, keeping other settings the same, we evaluate the frameworks without multi-direction scheme, 3DSE module, and ASP (i.e., using addition without distinction), respectively. The simulations are repeated for ten runs and the mean classification accuracies are reported in Table V, of which the best results are highlighted in bold. The up arrow means that the larger the value, the better the performance. From Table V, we can observe that MDN-ASP has the most outstanding performance and the accuracies degrade when any portion is discarded. The OAs of MDN-ASP significantly surpass MDN-ASP without multi-direction scheme, 3DSE module, and ASP by 1.54%, 2.39%, 1.26% on IP data set, respectively; by 2.02%, 0.81%, and 1.59% on UP data set, respectively; by 2.20%, 1.34%, and 1.29% on UH2013 data set, respectively. We can conclude that while our proposed framework looks rather complex, each component of it is fruitful and reasonable for achieving high-quality classification.

To further prove the generalization performance of the multi-direction scheme combined with ASP, which is denoted as MDASP here, we evaluate it on the most classic patch-based DL framework, that is, 3D-CNN networks. Specifically, experiments of 3D-CNN w/o MDASP and with MDASP are both carried out on the three data sets, and the average OA over ten executions are reported in Table VI. It is seen that the OAs of 3D-CNN with MDASP are obviously higher than that without MDASP. It indicates that the proposed MDASP is also of positive influence on 3D-CNN networks. Moreover, judging from the gain of accuracy in Tables V and VI, the multi-direction scheme and ASP play a more prominent role in UP and UH2013 than IP data set. This implies that the two components can effectively perceive and handle the image scene with many irregular class boundaries.

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TABLE V
CLASSIFICATION PERFORMANCE OF THE ABLATION STUDY

Dataset	Method	multi-direction	3DSE module	ASP	AA (%) ↑	OA (%) ↑	$k \times 100$ ↑
IP	MDN-ASP	✓	✓	✓	**93.99**	**94.55**	**93.09**
	w/o multi-direction		✓	✓	92.33	93.01	89.98
	w/o 3DSE module	✓		✓	93.94	92.16	91.02
	w/o ASP	✓	✓		92.68	93.29	91.73
UP	MDN-ASP	✓	✓	✓	**93.97**	**91.48**	**88.81**
	w/o multi-direction		✓	✓	91.58	89.46	87.34
	w/o 3DSE module	✓		✓	93.11	90.67	88.23
	w/o ASP	✓	✓		91.76	89.89	87.12
UH2013	MDN-ASP	✓	✓	✓	**90.52**	**86.98**	**85.89**
	w/o multi-direction		✓	✓	88.84	84.78	84.61
	w/o 3DSE module	✓		✓	89.39	85.64	84.91
	w/o ASP	✓	✓		89.75	85.69	85.10

TABLE VI
OVERALL ACCURACY (%) OF 3D-CNN W/O MDASP AND 3D-CNN WITH MDASP

Dataset	3D-CNN w/o MDASP	3D-CNN with MDASP
IP	81.67	**83.15**
UP	85.25	**87.46**
UH2013	79.80	**82.38**

D. Classification Performance

In this section, first, we compare the proposed MDN-ASP with the aforementioned classic and state-of-the-art methods. With respect to the detailed complement, SVM is cooperated with radial basis function (RBF) and the parameters are determined through five-fold cross-validation. The other algorithms are tested using the optimal configurations reported in the corresponding reference. Besides, for comparing the performance of different methods fairly, the spatial window size of 2D-CNN, 3D-CNN, SSRN, FDSSC, and the global region of DRCNN are set to $(2P-1)\times(2P-1)$, that is, 13×13 here. All algorithms for the three experimental data sets are executed up to ten times. Concretely, the average class-specified accuracies, AA, OA, and Kappa coefficient together with the standard deviation are shown in Tables VII–IX. The top-class record of each measurement (i.e., each row) is highlighted in bold.

From Table VII, we can find that the proposed MDN-ASP achieves the best results on IP data set. The PPF-CNN outperforms the SVM, 2D-CNN, and 3D-CNN with a large margin mainly benefited from the data augmentation strategy using pixel-pair features and majority voting inference. SSRN and FDSSCN obtain satisfactory results attributed to the advantages of skip connections, which can promote the generalization capability of the networks. In addition, profited by the intelligent utilization of the spatial contextual information, DR-CNN also acquires competitive performance. Nonetheless, as far as AA, OA, and Kappa coefficient, the proposed method outperforms SSRN by 1.23%, 2.37%, and 2.72%; FDSSCN by 0.15%, 3.99%, and 4.44%; DR-CNN by 4.03%, 3.98%, and 4.48%, respectively. It is worth mentioning that the standard deviation of MDN-ASP is the smallest among all approaches. Thus, it is reasonable to infer that our proposed framework has a preferable stability than other algorithms.

In Table VIII, some similar experimental results can be observed on the UP data set and the classification performance of our proposed MDN-ASP is superior to the comparisons. Although DR-CNN obtains remarkable accuracies, our proposed MDN-ASP outperforms it by a substantial margin, that is, 4.41% of AA, 1.53% of OA, and 2.03% of Kappa coefficient. It can be inferred that the multi-direction strategy together with the ASP has better capability to investigate the abundant spectral–spatial information included in the HSI. Besides, the continuous spectral and spatial 3DSERes modules ensure the network to extract more discriminative representations, which boost the classification performance.

Table IX presents the test results on the UH2013 image scene. We can see that the performance of almost all methods is inferior than that on the IP and UP data sets. It could be attributed to the scatter distribution of the training and test samples, which makes UH2013 a more challenging data set to develop potential spectral–spatial HSIC algorithms. Additionally, we can see that the proposed MDN-ASP reaches the highest level performance therein all models with respect to AA, OA, and Kappa coefficient, which again reveals the strength of our proposed multi-direction scheme and ASP module. Besides, the peak accuracies of the similar ninth and tenth classes (i.e., "Road" and "Highway") imply that the proposed method can precisely capture the distinguishing features to help recognize different materials with similar characteristics.

From visual perspective, Figs. 8–10 demonstrate the classification maps acquired by various algorithms using IP, UP, and UH2013 data sets, respectively, which are consistent to the quantitative results reported from Tables VII–IX. For convenient comparison, the corresponding false-color image and groundtruth are also exhibited together. Overall, compared with other methods, it is apparent that the proposed MDN-ASP generates more favorable maps with less misclassification. Due to the shortage of using spatial correlations, SVM undergoes serious scatter pepper-noise alike incorrectness on the segmentation maps. On the contrary, the pictures acquired by the other spectral–spatial classifiers are more compact, even over-smooth for some results, for example, Fig. 8(d)

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TABLE VII
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT ALGORITHMS FOR IP DATA SET

NO.	SVM [5]	2D-CNN [24]	3D-CNN [25]	PPF-CNN [29]	SSRN [34]	FDSSCN [35]	DR-CNN [46]	MDN-ASP
1	79.49	92.31	**100.00**	97.44	71.79	97.44	**100.00**	97.37
2	67.99	79.12	66.11	84.25	77.60	85.26	82.17	**93.52**
3	66.84	76.66	79.59	88.78	96.68	**99.11**	84.36	82.79
4	85.87	98.91	**100.00**	98.37	**100.00**	**100.00**	85.65	95.31
5	90.38	88.37	**97.99**	95.30	96.42	91.72	91.60	96.91
6	86.66	80.49	90.24	93.69	97.85	98.13	**100.00**	99.85
7	90.91	**100.00**	**100.00**	90.91	**100.00**	**100.00**	91.67	**100.00**
8	94.31	98.86	96.81	99.09	**100.00**	**100.00**	**100.00**	**100.00**
9	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
10	76.25	80.72	72.22	93.14	70.70	79.85	82.42	**94.25**
11	60.22	68.07	75.43	74.73	**96.69**	79.78	95.70	93.35
12	82.98	81.56	79.08	84.22	89.01	**98.05**	80.32	91.03
13	96.91	**100.00**	**100.00**	99.38	**100.00**	**100.00**	98.18	**100.00**
14	80.47	94.13	94.61	90.84	99.04	99.12	**99.66**	99.20
15	73.03	97.58	94.24	92.12	97.27	**99.70**	77.62	91.64
16	97.78	**100.00**	**100.00**	97.78	**100.00**	82.22	78.95	77.59
AA (%) ↑	83.13±0.67	89.80±1.14	90.40±1.03	92.50±1.19	93.32±0.87	94.40±0.93	90.52±0.71	**94.55±0.52**
OA (%) ↑	74.02±0.89	81.35±1.01	81.67±0.79	86.71±0.91	91.62±0.62	90.00±0.63	90.01±0.55	**93.99±0.31**
$k \times 100$ ↑	70.66±0.75	78.87±1.08	79.15±0.89	84.90±0.58	90.35±0.52	88.65±0.81	88.61±0.49	**93.09±0.22**

TABLE VIII
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT ALGORITHMS FOR UP DATA SET

NO.	SVM [5]	2D-CNN [24]	3D-CNN [25]	PPF-CNN [29]	SSRN [34]	FDSSCN [35]	DR-CNN [46]	MDN-ASP
1	84.23	72.59	82.61	94.15	95.11	98.81	98.62	**99.11**
2	66.43	98.16	81.54	83.66	77.19	75.06	**99.87**	99.11
3	68.43	55.59	57.19	52.95	43.14	74.55	84.51	**97.09**
4	97.87	96.50	98.73	94.88	89.32	96.57	96.40	**98.84**
5	99.37	99.82	99.55	99.82	**100.00**	98.02	98.92	**100.00**
6	92.43	54.24	92.98	97.48	**100.00**	98.05	67.99	61.64
7	89.91	54.54	81.86	93.58	**99.39**	84.00	91.16	98.99
8	92.60	92.87	97.24	99.41	**99.82**	95.54	71.44	90.92
9	97.36	92.33	94.21	**100.00**	99.12	96.86	97.17	**100.00**
AA (%) ↑	87.63±1.22	79.63±1.18	87.32±1.30	90.66±0.76	89.23±0.71	90.83±0.45	89.56±0.77	**93.97±0.40**
OA (%) ↑	78.89±1.16	85.48±1.15	85.25±1.44	88.66±1.15	85.48±1.14	85.99±1.57	89.95±0.83	**91.48±0.37**
$k \times 100$ ↑	73.41±1.51	79.98±1.25	80.77±1.41	85.15±1.22	81.22±1.65	81.98±1.39	86.78 ±1.05	**88.81±0.45**

TABLE IX
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT ALGORITHMS FOR UH2013 DATA SET

NO.	SVM [5]	2D-CNN [24]	3D-CNN [25]	PPF-CNN [29]	SSRN [34]	FDSSCN [35]	DR-CNN [46]	MDN-ASP
1	82.53	82.91	82.43	83.00	82.62	83.10	**99.54**	99.20
2	81.20	83.74	84.49	84.21	85.90	85.15	**99.56**	**99.56**
3	99.80	99.60	61.39	**100.00**	98.02	98.42	90.93	55.20
4	91.76	91.95	81.44	92.61	91.95	**99.81**	98.79	99.70
5	98.11	98.58	97.16	99.81	**100.00**	**100.00**	**100.00**	97.33
6	95.10	99.30	86.71	95.10	95.80	95.80	11.22	**100.00**
7	77.71	88.62	82.84	83.77	84.42	78.92	89.46	**94.47**
8	40.17	42.17	73.88	46.53	58.50	66.00	86.92	**92.57**
9	78.66	77.43	83.57	78.56	91.88	87.82	86.99	**95.04**
10	57.05	55.12	52.80	53.28	57.72	64.48	88.02	**96.03**
11	79.89	75.52	78.37	82.16	71.54	80.27	**89.02**	58.53
12	72.81	82.71	89.43	82.61	95.29	**98.94**	85.58	82.88
13	68.77	72.28	90.53	80.35	87.72	81.05	**96.28**	94.96
14	**100.00**	98.79	78.54	**100.00**	**100.00**	**100.00**	98.02	92.51
15	98.10	98.10	71.67	99.58	99.79	**100.00**	94.74	99.79
AA (%) ↑	81.44±1.21	83.12±0.87	79.68±1.03	84.11±0.66	86.74±0.54	87.98±0.78	87.67±0.57	**90.52±0.46**
OA (%) ↑	78.42±0.95	80.16±0.59	79.80±1.62	81.07±0.46	84.04±0.75	86.01±0.81	84.13±0.48	**86.98±0.28**
$k \times 100$ ↑	76.78±1.12	78.54±1.11	78.12±1.31	79.62±0.98	82.68±0.68	84.81±0.92	82.89± 0.69	**85.89±0.45**

of 2D-CNN and Fig. 8(h) of FDSSCN on IP data set. Furthermore, to clarify the capability of the proposed method to deal with the boundary pixels, the border regions of interest (BRoI) are highlighted in black circles for IP and UP data sets. We can easily find that the BRoIs of the patch-wise spectral–spatial classifiers, that is, 3D-CNN, SSRN, FDSSCN, and DR-CNN, are quite confusing and inaccurate, for the spatial context information is deficient to these boundary pixels. In contrast,

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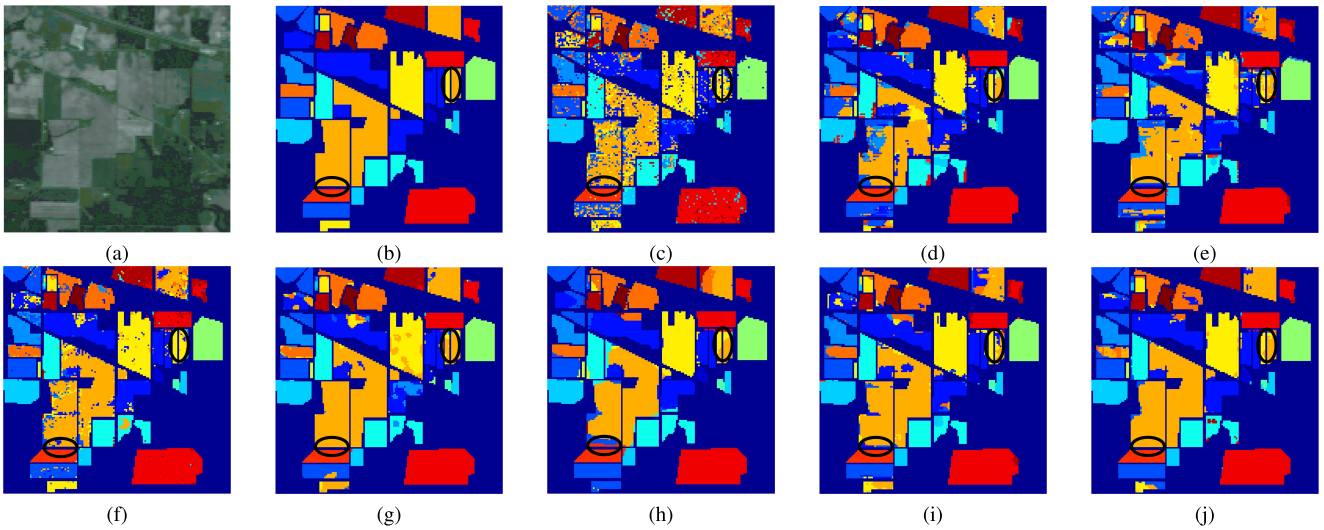

Fig. 8. Classification maps obtained via various algorithms for IP data set. (a) False-color image. (b) Groundtruth. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) PPF-CNN. (g) SSRN. (h) FDSSCN. (i) DR-CNN. (j) MDN-ASP. Best zoomed-in view.

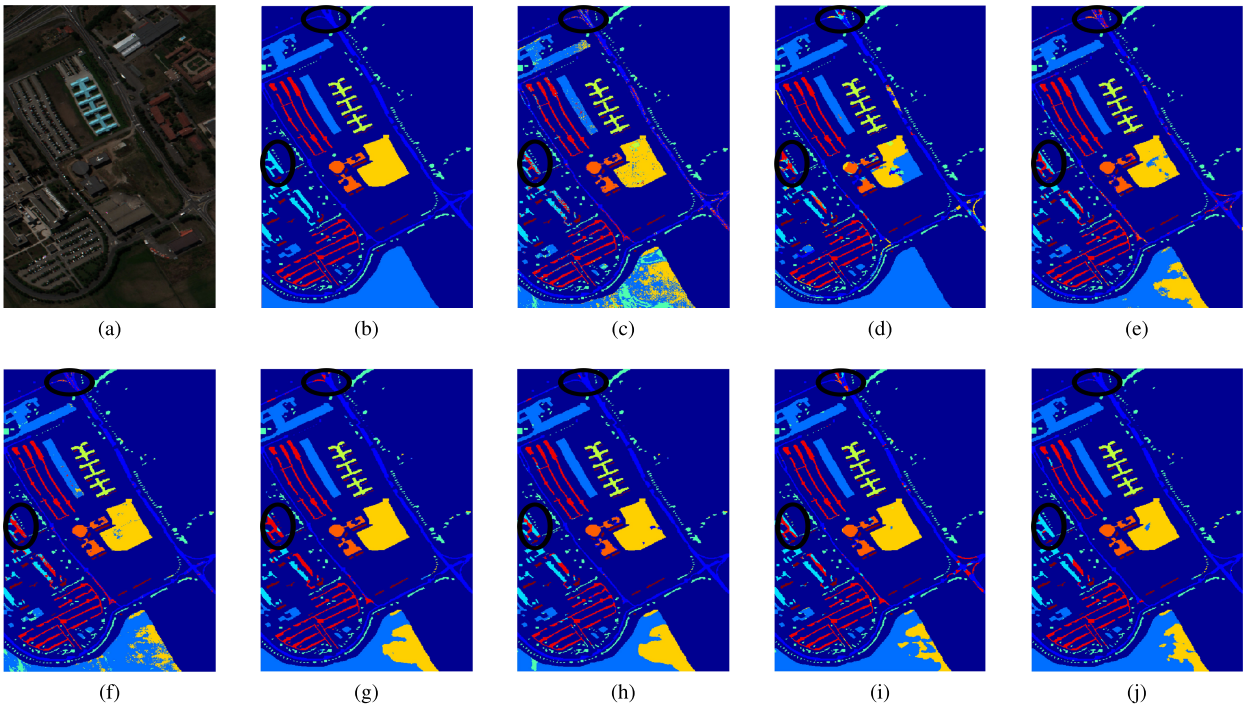

Fig. 9. Classification maps obtained via various algorithms for UP data set. (a) False-color image. (b) Groundtruth. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) PPF-CNN. (g) SSRN. (h) FDSSCN. (i) DR-CNN. (j) MDN-ASP. Best zoomed-in view.

our proposed MDN-ASP is able to generate comparative elegant and clear classification results [see Figs. 8(j) and 9(j)] on these difficult BRoIs. This further proves the advantages of our proposed multi-direction strategy, ASP, and successive 3DSERes blocks to thoroughly and precisely exploit the spectral–spatial features of each sample.

Particularly, apart from the previous experiments, to further validate the advancement of the MDN-ASP, two other recently proposed multi-stream ensemble networks are employed as baseline approaches. They are also constituted in a multi-branch manner, where the extracted deep features are fused before being imported into the subsequent classifier. One is the CNN-based multilayer spatial–spectral feature fusion and sample augmentation with local and nonlocal constraints, which is denoted as MSLN-CNN [44]. Another is the multiscale dynamic graph convolutional network abbreviated as MDGCN [53]. We utilize all experimental data sets to make the comparisons and the training sample partitioning are kept same to Tables I–III. The mean classification accuracies over ten runs are reported in Table X. It is observed that MDN-ASP can produce improved performance on all data sets than the baseline methods (only except for the AA of MDGCN on IP

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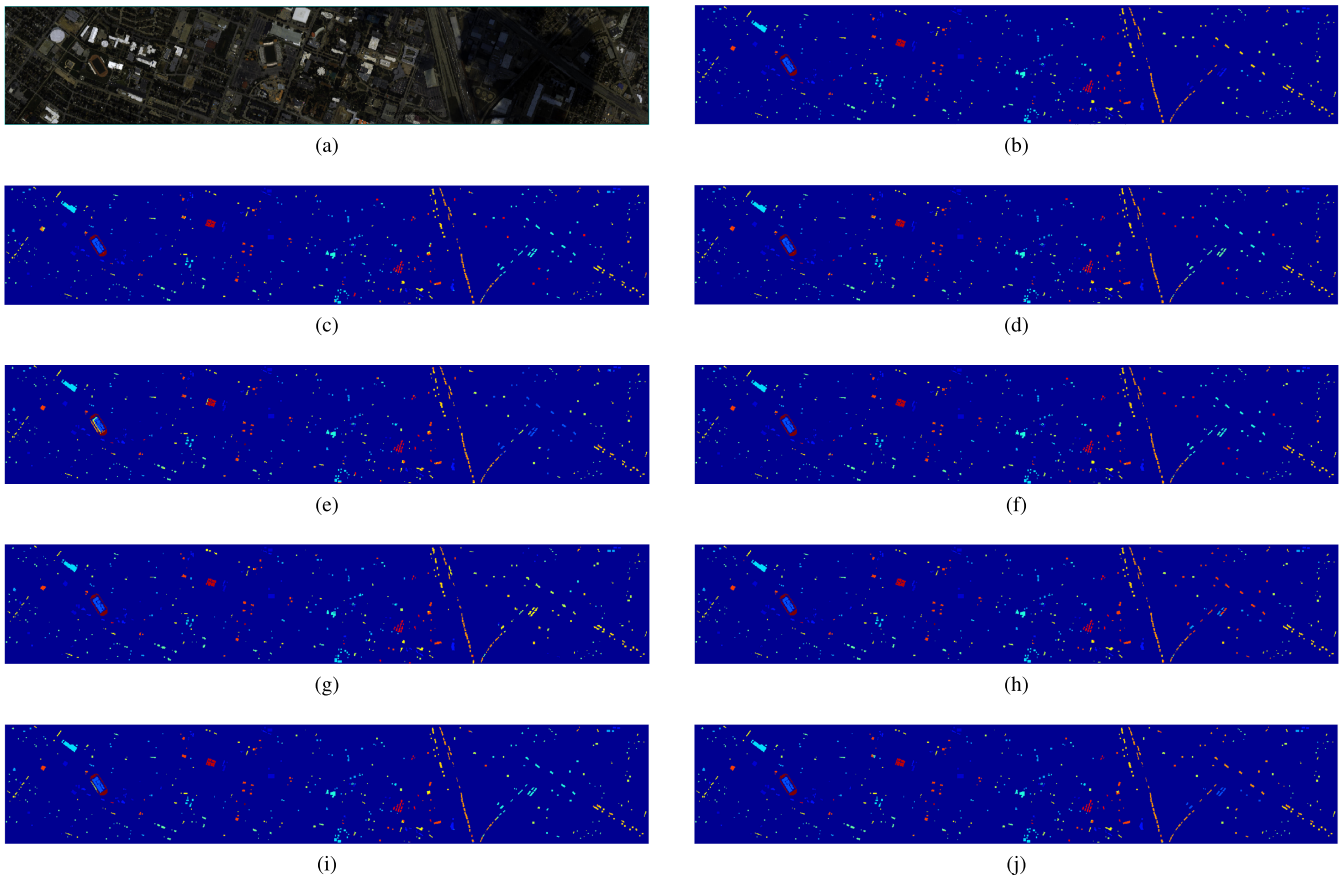

Fig. 10. Classification maps obtained via various algorithms for UH2013 data set. (a) False-color image. (b) Groundtruth. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) PPF-CNN. (g) SSRN. (h) FDSSCN. (i) DR-CNN. (j) MDN-ASP. Best zoomed-in view.

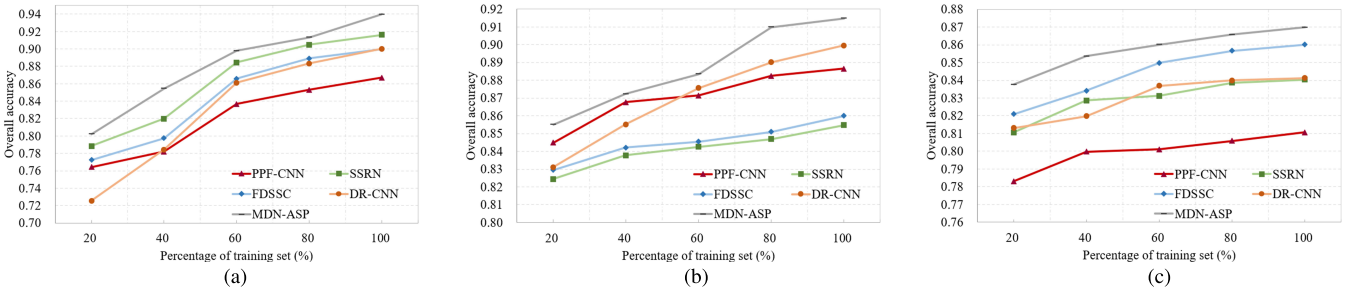

Fig. 11. Overall accuracies of PPF-CNN, SSRN, FDSSCN, DR-CNN, and MDN-ASP with different percentages of training set. (a) IP. (b) UP. (c) UH2013.

and the OA of MDGCN on UP), especially for the challenging UH2013 data set, which can convincingly prove the superiority of the proposed MDN-ASP.

E. Influence of Different Number of Training Samples

In real-world applications, labeling the land-cover type of each pixel is time-consuming and labor-intensive, thus the training samples are generally insufficient. Therefore, we explore the classification performance versus fewer training samples, that is, different portions of the training data from 20% to 100%. At this point, 100% refers right to the full training samples, as shown in Tables I–III. As for other percentages, the training samples are randomly selected from the full training data to ensure more reliable results. The experiments are repeated ten times, and then the average OA of the most competitive compared methods, that is, PPF-CNN, SSRN, FDSSCN, DR-CNN, and our proposed MDN-ASP are depicted in Fig. 11. From Fig. 11, it is observed that the overall accuracies are improved as the labeled samples increasing for all algorithms, and our proposed MDN-ASP ranks the first place all the time. Particularly, on the most challenging classification data set UH2013, MDN-ASP obtains approximately 84% overall classification accuracy with only 20% training samples, which sufficiently guarantees the effectiveness of the proposed method under severe small sample size conditions.

F. Analysis of Neighborhood Size

To assess the impact of different spatial context scopes, we test our proposed method with input of varying

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TABLE X
CLASSIFICATION ACCURACIES OF MDN-ASP AND THE RECENT BASELINE MULTI-STREAM NETWORKS ON THE EXPERIMENTAL DATA SETS

Dataset	Method	AA (%) ↑	OA (%) ↑	$k \times 100$ ↑
IP	MSLN-CNN	89.82	91.12	89.83
	MDGCN	**96.08**	92.12	90.01
	MDN-ASP	93.99	**94.55**	**93.09**
UP	MSLN-CNN	90.14	87.57	83.35
	MDGCN	91.27	**91.65**	88.68
	MDN-ASP	**93.97**	91.48	**88.81**
UH2013	MSLN-CNN	82.81	80.52	78.87
	MDGCN	83.00	81.00	79.53
	MDN-ASP	**90.52**	**86.98**	**85.89**

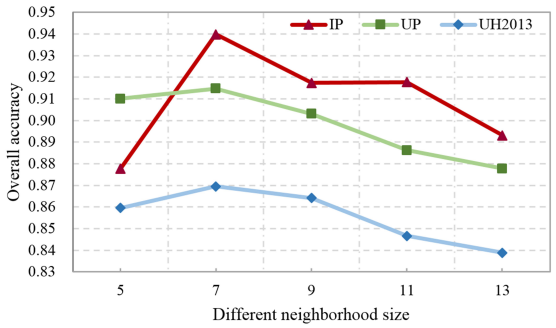

Fig. 12. Overall accuracies with different neighborhood size for three experimental data sets.

neighborhood size, which is set to an odd number ranging from 5 to 11. The overall classification accuracies are delineated in Fig. 12. It is observed that the optimal performance is achieved when the window size equals to 7. Most obviously, the red polyline marked by triangles of IP data set shows that the OA obtains noticeable increment when the patch size turns from 5 to 7, but as the window getting bigger, it drops off. The reason is when the patch includes more neighboring pixels, it has increasing probability to contain materials from different categories, notably for the border areas across various land-cover types. Then, the benefits of the proposed multi-direction strategy and ASP will be degraded to some extent. As a result, the spatial boundaries of different classes are not perfectly preserved and the classification maps are excessively smooth. Additionally, it is a fact that more calculation and hardware resources are required with larger neighborhood size, thereby specifying it to 7 is a feasible alternative.

G. Computational Efficiency

In this section, we examine the computational complexity of our proposed method. The elapsed training and test time of the presented MDN-ASP, and the competitive patch-wise classification models, that is, SSRN, FDSSCN, and DR-CNN are summarized in Table XI. It is observed that FDSSCN and SSRN are faster than the other two methods since the network is narrow with simple single patch input. DR-CNN is the most time-consuming due to the pre-training process of each

TABLE XI
ELAPSED TIME (IN S) OF THE TRAINING AND TEST PROCEDURE OF DIFFERENT METHODS

Method		IP	UP	UH2013
SSRN	Training	389.13	725.94	580.07
	Test	5.21	12.09	6.45
FDSSCN	Training	366.70	534.54	512.02
	Test	9.65	18.46	7.48
DR-CNN	Training	848.37	2394.79	2580.14
	Test	345.35	105.58	293.95
MDN-ASP	Training	725.41	2855.38	2285.75
	Test	21.63	48.18	19.94

sub-network in the training stage. The proposed MDN-ASP also takes a long time owing to the calculation burden of the multi-direction patches, but it is acceptable as obtaining satisfactory classification performance. Meanwhile, the framework of MDN-ASP is regular with the same architecture in each stream, thus the network has the potential to utilize multiple GPUs for parallel computing, which can greatly reduce the running time.

IV. CONCLUSION

In this article, a novel MDN cooperated with ASP (MDN-ASP) is proposed for HSI classification. The designed MDN comprises nine paralleled channels to fully investigate the wealthy spectral–spatial information contained in the HSI through multi-direction samples. Additionally, taking the image-spectrum merged structure of the HSI into account, we devise successive spectral and spatial 3DSERes modules to emphasize the significant discriminative features and suppress the trivial ones in each tunnel of the networks, which can further boost the classification performance. Subsequently, in order to flexibly aggregate the extracted multi-direction deep spectral–spatial features, the simple yet effective ASP is involved into the networks by reusing the natural prior spectral signatures, which is physically interpretable. The ablation studies demonstrate that this architecture design is reasonable to promote the classification performance, even under the condition of small sample size. Furthermore, the experiment results on three benchmarks also indicate that our proposed framework can attain superior outcomes than the classic and SOTA algorithms, especially for the difficult boundary pixels.

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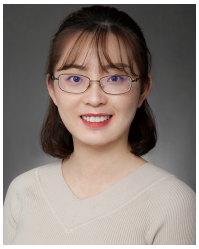

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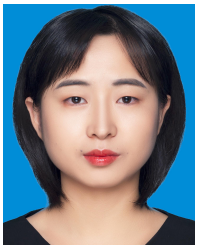

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